To Infer or not to Infer? Natural Language Inference and Computational Semantics

Stergios Chatzikyriakidis
Centre for Linguistic Theory and Studies in Probability (CLASP),
FLoV

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Outline

1. Intro
2. Three Eras of NLI
3. Brittleness and Generalizability
4. Where Do we Go From Here?

Natural Language Inference

- Roughly: the task of determining whether a NL hypothesis H follows from an NL premise P
  - 'inferential ability is not only a central manifestation of semantic competence but is in fact centrally constitutive of it' Cooper et al. (1996)
  - inferential ability as a means to test the semantic adequacy of NLP systems

What Humans Infer?

- Reasoning: part of our every day routine:
  - we hear Natural Language (NL) sentences
  - we participate in dialogues
  - we read books or legal documents.
- Successfully understanding, participating or communicating with others in these situations presupposes some form of reasoning
  - about individual sentences
  - whole paragraphs of legal documents
  - small or bigger pieces of dialogue

Different domains, different reasoning

- The variety of reasoning is difficult to be explained by a single coherent system of reasoning. Why?
  - because reasoning is performed in different ways in each one of them

Consider the following example:

(1) Three representatives are needed.

Different domains, different reasoning

(2) Three representatives are needed.

- Assume a human reasoner with expert knowledge in a legal context:
  - s/he will most probably judge that a situation where more than three representatives are provided could be compatible with the semantics of the utterance
- The same reasoner interpreting the utterance as part of a casual, everyday conversation:
  - *three* would most likely be interpreted as *exactly three*

Incremental Reasoning in Interaction

- Reasoning can get very complicated as soon as we move to a dialogue setting:
  - A. Mont Blanc is higher than
  - B. Mt. Ararat?
  - A. Yes.
  - B. No, this is not correct. It is the other way around.
  - A. Are you...
  - B. Sure? Yes, I am.
  - A. Ok, then.

Inference can be Incremental and Interactive

- The listener reasons based on:
  - utterances that are split between two participants
    - thus having to dynamically keep track of them.
- Furthermore, the listener must be able to compute:
  - global inferences, i.e. inferences that are based on statements/facts that are shared (agreed upon) by the dialogue participants
    - local inferences that are based on facts that are not shared by all dialogue participants.

Can machines take it?

- The grand picture, the mother of all tasks: create computational semantics systems that will reflect this wealth of reasoning patterns
  - Start small: systems that perform reasoning on well-defined (well...) subsets of what NLI is
    - Define the task
    - Provide the system
    - Evaluate the system

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The Three Eras of NLI

- The Symbolic Era
  - What we need is a solid formal system and good rules formalized in that system
  - We just have to find these good rules
    - It should be possible, no?

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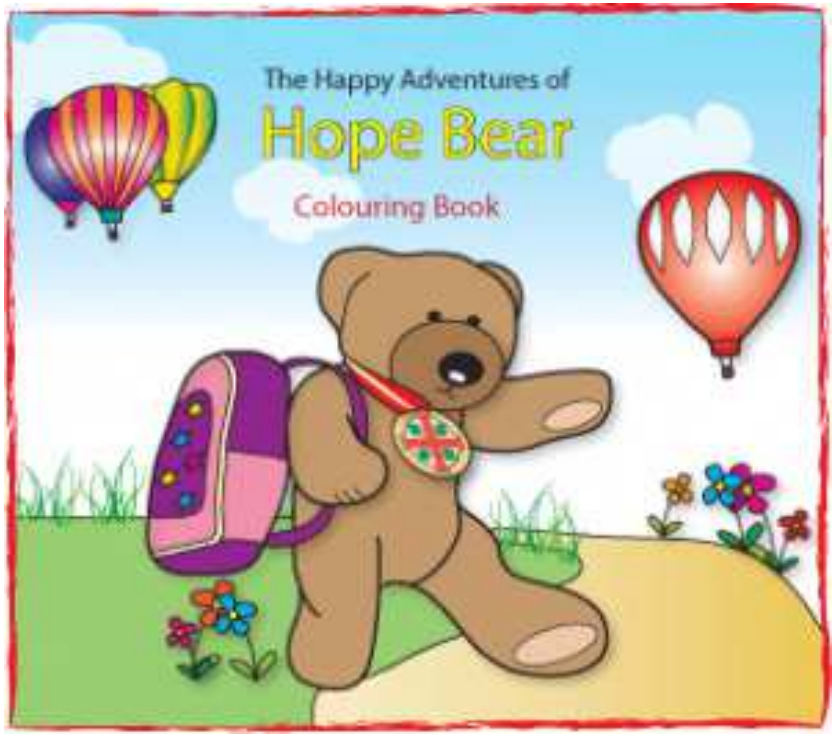

The Symbolic Era: the Datasets

- The FraCas test suite Cooper et al. (1996)
  - a collection of mostly logical entailments. Categorization is done according to semantic category
  - Three way classification of 346 inference problems: YES (the conclusion follows), NO (the negation of the conclusion follows) and UNK (none of the two follow)

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The Symbolic Era: the Datasets

(3) A Swede won the Nobel Prize.
Every Swede is Scandinavian.
Did a Scandinavian win the Nobel prize? [Yes, FraCas 049]

(4) No delegate finished the report on time..
Did any Scandinavian delegate finish the report on time?
[No, FraCaS 070]

The Symbolic Era: the Datasets

(5) A Scandinavian won the Nobel Prize.
Every Swede is Scandinavian.
Did a Swede win the Nobel prize? [UNK, FraCaS 065]

- Other typical examples

(6) Either Smith, Jones or Anderson signed the contract. Did John sign the contract? [UNK] (plurals, FraCaS 083)

(7) Dumbo is a large animal. Is Dumbo a small animal? [NO] (adjectives, FraCaS 205)

(8) Smith believed that ITEL had won the contract in 1992. Did ITEL win the contract in 1992? [UNK] (Attitudes, FraCaS 334)

The Symbolic Era: The General Approach

- Pair a symbolic syntactic parser with logical semantics
  - Define a correspondence between abstract syntax (usually abstract syntactic trees) and semantics in some logical language

The Symbolic Era

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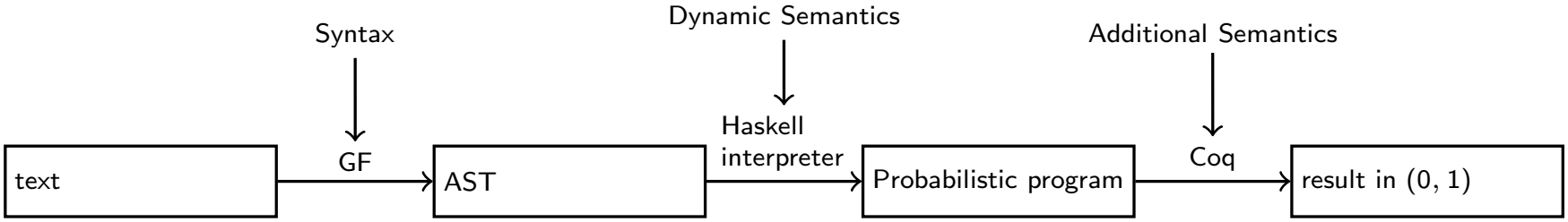
The Symbolic Era: The State of the Art

- Bernardy and Chatzikyriakidis (2019)
  - Building on earlier work Bernardy and Chatzikyriakidis (2017)
- In a nutchell:
  - A converter from syntax trees to types.
    - GF syntax trees to Coq types via a Haskell Program, using Monadic Dynamic Semantics
  - Type-theoretical combinators for the treatment of various linguistic phenomena (e.g. adjectives)

The Symbolic Era: The State of the Art

- DFraCoq's architecture



The Symbolic Era: The State of the Art

- Covers more of the FraCaS than any previous account (around 80%)
  - First run on the ellipsis and anaphora section
  - Overall accuracy of around 89.2%

The Symbolic Era: The State of the Art

- Comparison of DFraCoq with previous logical approaches

Section	#cases	Ours	FC	MINE	Nut	Langpro
Quantifiers	75	.96	.96	.77	.53	.93
		(74)				(44)
Plurals	33	.82	.76	.67	.52	.73
						(24)
Anaphora	28	.86	-	-	-	-
Ellipsis	52	.87	-	-	-	-
Adjectives	22	.95	.95	.68	.32	.73
		(20)				(12)
Comparatives	31	.87	.56	.48	.45	-
Temporal	75	-	-	-	-	-
Verbs	8	.75	-	-	-	-
Attitudes	13	.92	.85	.77	.46	.92
						(9)
Total	337	.89	.83	.69	.50	.85
		(259)	(174)	(174)	(174)	(89)

The Symbolic Era: The State of the Art

- High precision, but low recall when thrown into open text
  - Brittle systems

The Classic Machine Learning Era

- Late 90s, all the way to approximately 2013-2014
  - Supervised Machine Learning: using hand-crafted features to train the models
    - Bag-of-Words, similarity metric based models, Maximum Entropy Classifiers, SVMs, Naive Bayes... the usual suspects
    - Enough with Logic, this will definitely work

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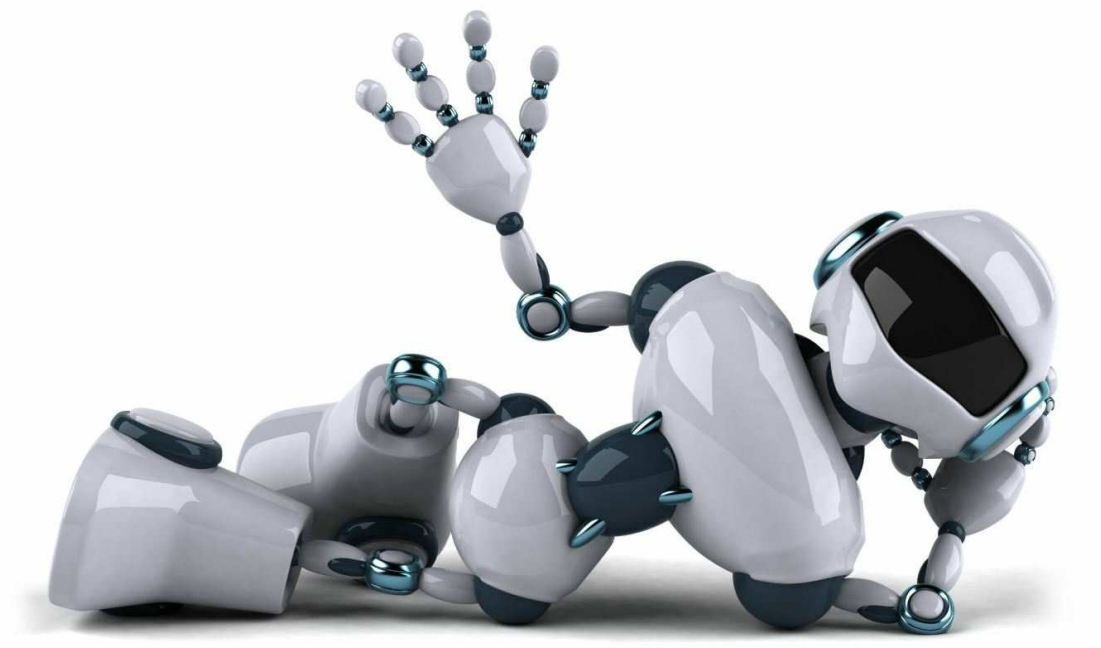

The Classic Machine Learning Era: Datasets

- The PASCAL Recognizing Textual Entailment Challenges (RTE)

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Some RTE examples

P Budapest again became the focus of national political drama in the late 1980s, when Hungary led the reform movement in eastern Europe that broke the communist monopoly on political power and ushered in the possibility of multiparty politics.

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P Like the United States, U.N. officials are also dismayed that Aristide killed a conference called by Prime Minister Robert Malval in Port-au-Prince in hopes of bringing all the feuding parties together.

H U.N. officials take part in a conference called by Prime Minister Robert Malval. (does not follow, RTE1933)

The RTE datasets

- Performance on the RTE datasets has been in general poor
  - Around 60% accuracy for many years and thoughout the challenges

The Deep Learning Era

- The dominant (almost the only one) nowadays
  - Some sort of Artificial Neural Network model is used to deal with NLI
  - ANNs are great learners, they can approximate any function given enough data (oversimplified, for a precise definition, check Universal Approximation Theorem)
  - But enough is most of the times a lot!
    - And quality might not be great

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The Deep Learning Era

- New datasets are needed
  - Both FraCaS and the RTE datasets (also SICK) are not suitable on datasize alone

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The SNLI platform

- Instructions used on Mechanical Turk

We will show you the caption for a photo. We will not show you the photo. Using only the caption and what you know about the world:

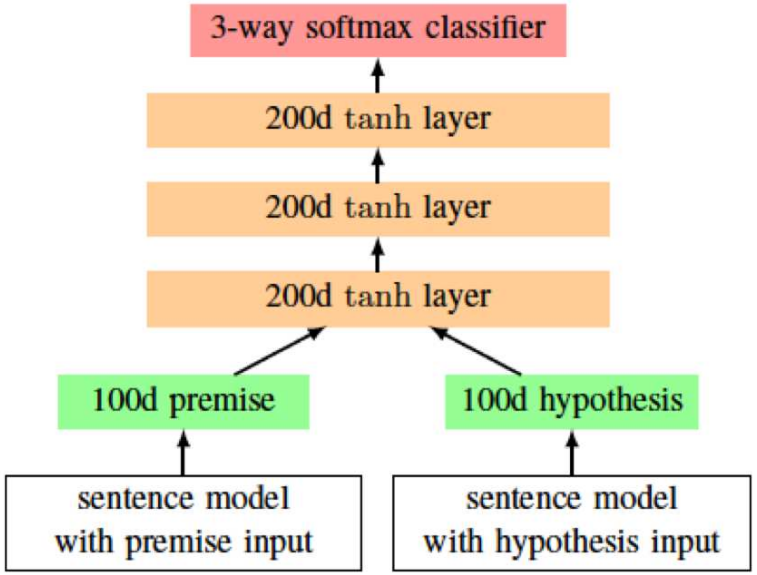
- Write one alternate caption that is **definitely** a **true** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "There are animals outdoors."*
- Write one alternate caption that **might be a true** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "Some puppies are running to catch a stick."*
- Write one alternate caption that is **definitely** a **false** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "The pets are sitting on a couch." This is different from the* maybe correct *category because it's impossible for the dogs to be both running and sitting.*

Impressive results

- The State of the Art at this moment on the SNLI dataset is above 90% accuracy
  - Actually, the State of the Art has been surpassed tenths of times, since the first paper using NNs for SNLI (Bowman et al., 2015)

The original approach

- Simple architecture
  - three 200d tanh layers plus a bottom layer
    - input: concatenated sentence representations
    - output: softmax classifier (3-way)
    - Achieves an accuracy close to 78%



The BERT subera

- Tenths of systems since then, slowly advancing the SoA by proposing different NN architectures and network tweaks
- Where we are now: the BERT era (Bidirectional Encoder Representations from Transformers)
  - What it does?
    - Transfer learning: transfer knowledge obtained from one task to another (e.g. semantic similarity to NLI)
    - Pre-training: pre-train the model on two prediction tasks (one is e.g. predict the next sentence)
    - Fine-tuning for other tasks: no training from scratch is needed, just fine-tuning of parameters

The BERT subera

- Impressive in a wide variety of tasks (basically all of the GLUE benchmarks)

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	**Average** -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.9	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	88.1	91.3	45.4	80.0	82.3	56.0	75.2
$\text{BERT}_{\text{BASE}}$	84.6/83.4	71.2	90.1	93.5	52.1	85.8	88.9	66.4	79.6
$\text{BERT}_{\text{LARGE}}$	**86.7/85.9**	**72.1**	**91.1**	**94.9**	**60.5**	**86.5**	**89.3**	**70.1**	**81.9**

Table 1: GLUE Test results, scored by the GLUE evaluation server. The number below each task denotes the number of training examples. The "Average" column is slightly different than the official GLUE score, since we exclude the problematic WNLI set. OpenAI GPT = (L=12, H=768, A=12); $\text{BERT}_{\text{BASE}}$ = (L=12, H=768, A=12); $\text{BERT}_{\text{LARGE}}$ = (L=24, H=1024, A=16). BERT and OpenAI GPT are single-model, single task. All results obtained from https://gluebenchmark.com/leaderboard and https://blog.openai.com/language-unsupervised/.

- Current SoTA: BERT + Semantic Role Labelling information (Zhang et al., 2019)

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Brittleness through the back door?

- At first sight, NN systems do not seem to be suffering from the brittleness problem
  - Only to some extent correct
  - Recent studies seem to propose that NN systems are also brittle, but in another sense
    - They fail to generalize outside individual datasets and are, furthermore, unable to capture certain NLI patterns, at all

Brittleness through the back door?

- NLI systems have limited generalization ability outside the datasets that they are trained and tested on Glockner et al. (2018)
  - NLI systems break easily when, instead of being tested on the original SNLI test set, they are tested on a test set which contain sentences that differ by at most one word from sentences in the training set
    - Significant drop in accuracy, e.g. between 22 and 33 points when trained on SNLI and tested on the new dataset, is reported for three out of four state-of-the-art systems tested.
    - The system less prone to breaking is Kim et al. (2018) (5 points drop when trained on SNLI and tested on the new dataset), which utilizes external knowledge taken from WordNet Miller (1995).

Brittleness through the back door?

- NLI systems have limited generalization ability outside the datasets that they are trained and tested on (Talman and Chatzikyriakidis, 2018)
  - Train and test six state-of-the-art NN models using train and test sets drawn from a different corpus
    - E.g. the train set is drawn from the SNLI but the test from the MultiNLI, vice versa and other similar combinations
  - The results shows an average drop of 24.9 points in accuracy for all systems, including the system by Kim et al. (2018)

How our NLI systems are doing: brittleness through the backdoor

Train	Dev	Test	Test Accuracy	Delta	Model
SNLI	**SNLI**	**SNLI**	**86.1**		600D BiLSTM-max
SNLI	**SNLI**	**SNLI**	**86.6**		600D HBMP Talman et al. (2018)
SNLI	**SNLI**	**SNLI**	**88.0**		600D ESIM Chen et al. (2017)
SNLI	**SNLI**	**SNLI**	**88.6**		300D KIM Kim et al. (2018)
SNLI	SNLI	MultiNLI-m	55.7*	-30.4	600D BiLSTM-max
SNLI	SNLI	MultiNLI-m	56.3*	-30.3	600D HBMP
SNLI	SNLI	MultiNLI-m	59.2*	-28.8	600D ESIM
SNLI	SNLI	MultiNLI-m	61.7*	-26.9	300D KIM
SNLI	SNLI	SICK	54.5	-31.6	600D BiLSTM-max
SNLI	SNLI	SICK	53.1	-33.5	600D HBMP
SNLI	SNLI	SICK	54.3	-33.7	600D ESIM
SNLI	SNLI	SICK	55.8	-32.8	300D KIM
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**73.1***		**600D BiLSTM-max**
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**73.2***		**600D HBMP**
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**76.8***		**600D ESIM**
MultiNLI	**MultiNLI-m**	**MultiNLI-m**	**77.3***		**300D KIM**
MultiNLI	MultiNLI-m	SNLI	63.8	-9.3	600D BiLSTM-max
MultiNLI	MultiNLI-m	SNLI	65.3	-7.9	600D HBMP
MultiNLI	MultiNLI-m	SNLI	66.4	-10.4	600D ESIM
MultiNLI	MultiNLI-m	SNLI	68.5	-8.8	300D KIM
MultiNLI	MultiNLI-m	SICK	54.1	-19.0	600D BiLSTM-max
MultiNLI	MultiNLI-m	SICK	54.1	-19.1	600D HBMP
MultiNLI	MultiNLI-m	SICK	47.9	-28.9	600D ESIM
MultiNLI	MultiNLI-m	SICK	50.9	-26.4	300D KIM
SNLI+MultiNLI	**SNLI**	**SNLI**	**86.1**		**600D BiLSTM-max**
SNLI+MultiNLI	**SNLI**	**SNLI**	**86.1**		**600D HBMP**
SNLI+MultiNLI	**SNLI**	**SNLI**	**87.5**		**600D ESIM**
SNLI+MultiNLI	**SNLI**	**SNLI**	**86.2**		**300D KIM**
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SNLI+MultiNLI	SNLI	SICK	54.5	-33.0	600D ESIM
SNLI+MultiNLI	SNLI	SICK	54.6	-31.6	300D KIM

- Test accuracies (%). For results highlighted in bold the training data include examples from the same corpus as the test data. For the other cases, the training and test data involve separate corpora.

The issue of generalizability

- BERT seems to be doing much better than the other models in our examples
  - Did not exist when the other negative results papers appeared
- Still when moved from more similar to less similar datasets (this is to be expected, of course, though the drop is huge)

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Where do we go from here?

- Keep all research directions open
  - There seems to be still use for research directions that were deemed useless in the field
- Attempt integration
  - Hybrid approaches
    - Symbolic + DL: On what level, what are the gains if at all
  - Get a clear idea of what approach works well with what, but most importantly, what approach DOES NOT work well with what

Where do we go from here? Better datasets

- Reflect in a more accurate way the range of inference patterns found in human reasoning with NL
  - Most importantly: we need datasets that can test NLP systems on reasoning with dialogue
- Probabilistic datasets?
  - NLI datasets that give probability scores for an inference instead of a three way classification
  - Needed in order to test probabilistic semantics systems like the ones currently developed at CLASP (e.g. Cooper et al. (2015); Bernardy et al. (2018)

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